

The Fermionic $U(n)^3$ Quiver: BPS Entropy, Concentration, and Chaos

Progress report, October 2026

Abstract. We study a purely fermionic, $\mathcal{N} = 2$ supersymmetric gauged quantum mechanics that, as far as we can tell, has not been considered before: three bifundamental fermions A, B, C of $U(n)^3$ arranged on a triangle quiver, each in p flavours, with the cubic supercharge $Q = \sum C_{abc} \text{Tr}(A^a B^b C^c)$. It is the matrix ($D = 2$) member of the coloured models of Gurau–Witten type, so its large- n limit is planar, and it was chosen because bifundamental matter has no zero weights, which removes the mechanism that makes the BPS window of adjoint matrix models grow with the rank. In the gauge-singlet sector, exact cohomology at $(n, p) = (2, 2)$ and Lanczos diagonalisation at $(2, 3)$ show that all singlet BPS states, 90 and 1680 of them, sit at half filling, and all 90 classes at $(2, 2)$ are fortuitous. A Schur–Weyl singlet basis extends concentration to $(3, 2)$, and a semidefinite bootstrap of the singlet sector, validated against all exact data, certifies that the $(3, 3)$ model has no singlet BPS states in degrees $k \leq 21$, which makes all 1680 classes at $(2, 3)$ fortuitous as well. At $p = 3$ the singlet multiplet spectrum has Gaussian-orthogonal level statistics over 10024 levels and small gaps only next to the BPS degree, while $p = 2$ turns out to be special, with a hidden $\mathcal{N} = 4$ supersymmetry on singlets and a spectrum that is not random-matrix like. We then show that the singlet index is exactly the partition function of a three-species log gas on the circle. This proves $I_0(n, 2) = (-1)^n (3n)! / (n!)^3$ and fixes the sign of the index in general, and the large- n saddle gives $\ln |I_0| = F^*(p) n^2 + O(\ln n)$ with $F^* > 0$ exactly when $p > 2$; at $p = 3$, $F^* = 1.2373$, which is 41% of the singlet log-count, and as $p \rightarrow \infty$ the BPS entropy per fermion approaches the $\mathcal{N} = 2$ SYK value $\frac{1}{2} \ln 3$. A parameter-free finite- n evaluation of the same saddle reproduces all 27 non-zero exact indices and singlet counts to within 0.6 in the logarithm. The $p \geq 3$ quiver therefore has macroscopic BPS entropy, concentration, fortuity and chaos at the sizes we can reach; concentration near half filling at $(3, 3)$ and beyond, and genuine near-BPS scaling, are the open questions, and we describe how to attack them.

1 Introduction

The fortuity programme proposes that the BPS states responsible for the entropy of supersymmetric black holes are cohomology classes that exist at a given rank but cannot be continued to larger rank, as opposed to monotone classes that are inherited along a tower of ranks [1]. Chang, Chen, Sia and Yang argued that a theory whose BPS states are mostly fortuitous should exhibit R-charge concentration [2]: within each complex, the BPS states sit in a single degree of the R-charge. $\mathcal{N} = 2$ SYK is the paradigm [3]. There the BPS states occupy a window of width $\hat{q}$ around half filling, their number grows exponentially with the number of fermions, and the non-BPS spectrum next to them is chaotic, with the density of states of the $\mathcal{N} = 2$ super-Schwarzian [4]. Chen asked for the simplest *matrix* model with these properties, a “matrix SYK” model: purely fermionic matrix quantum mechanics with a cubic supercharge built from traces [5]. He showed that a single adjoint fermion already has fortuitous BPS states, although they spread over $N + 1$ charges, and suggested that three adjoint flavours might concentrate.

Our earlier reports [6, 7] analysed this adjoint family in detail and reached a negative conclusion. At the maximal weight the BPS window has width $W = r(w_s - 1) + 1$ for a gauge algebra of rank r , where the width $w_s \geq 2$ contributed by a single Cartan site is controlled by one bilinear form. The window therefore grows with the rank, and a concentrated large- N limit is impossible for adjoint matter at fixed degree of the supercharge. The mechanism lives on the Cartan: the adjoint representation has zero weights, and at the maximal weight the complex is a product of Cartan sites, each of which contributes its own spread in charge. This points to an obvious way out, namely to build the cubic supercharge from matter without zero weights. The simplest such matter is bifundamental, and the simplest cubic gauge invariant of bifundamentals is the trace around a triangle, $\text{Tr}(ABC)$.

This report collects what we have learned about the resulting model. We describe it in Section 2 and place it in the literature in Section 3. Section 4 presents the exact and numerical results on BPS states at small rank: singlet indices, concentration at $(n, p) = (2, 2)$ and $(2, 3)$, and a fortuity test. Section 5 turns to the non-BPS singlet spectrum, where $p = 3$ is chaotic and $p = 2$ has a hidden extended supersymmetry. Section 6 contains the main analytic result: the singlet index is the partition function of a three-species log gas, which we solve at large n . Section 7 goes beyond $n = 2$, with a Schur–Weyl singlet basis at $(3, 2)$ and a semidefinite bootstrap of the singlet sector at $(3, 3)$. Section 8 assesses the prospects of the $p \geq 3$ model and lists the next steps.

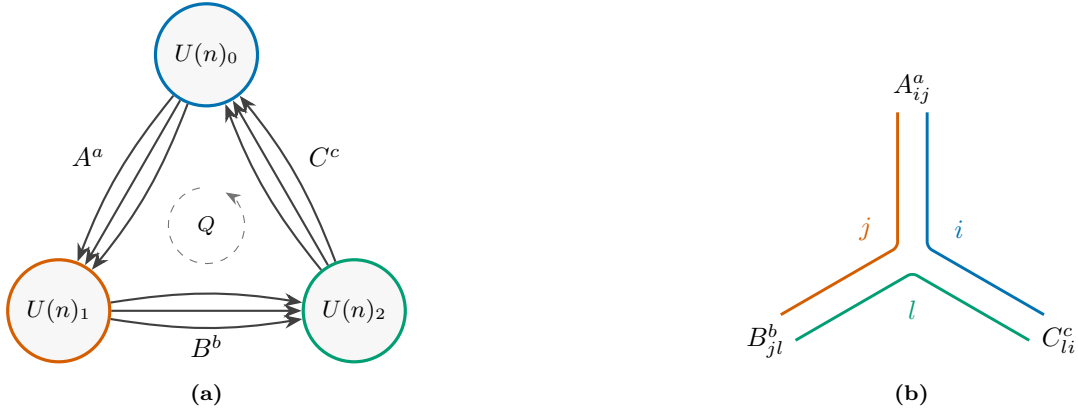


Figure 1: (a) The quiver. An arrow from node v to node $v + 1 \pmod{3}$ is a bundle of p fermionic bifundamentals in $(\mathbf{n}_v, \bar{\mathbf{n}}_{v+1})$, drawn here for $p = 3$. The supercharge (2) is the gauge-invariant loop around the triangle, with a general flavour tensor C_{abc} . (b) The interaction as a ribbon vertex. Each field carries the indices of the two nodes it joins, and each index line closes around the vertex. This is the $D = 2$ member of the coloured models of [8]; at large n only planar ribbon graphs survive.

2 The model

The gauge group is $U(n)_0 \times U(n)_1 \times U(n)_2$, with fundamental indices i, j and l for the three nodes. The matter consists of complex fermions

$$A^a_{ij} \in (\mathbf{n}, \bar{\mathbf{n}}, \mathbf{1}), \quad B^a_{jl} \in (\mathbf{1}, \mathbf{n}, \bar{\mathbf{n}}), \quad C^a_{li} \in (\bar{\mathbf{n}}, \mathbf{1}, \mathbf{n}), \quad a = 1, \dots, p, \quad (1)$$

transforming as $A \rightarrow g_0 A g_1^{-1}$, $B \rightarrow g_1 B g_2^{-1}$ and $C \rightarrow g_2 C g_0^{-1}$, with canonical anticommutators such as $\{A^a_{ij}, (A^b_{i'j'})^\dagger\} = \delta^{ab} \delta_{ii'} \delta_{jj'}$. We regard A, B and C as creation operators acting on a Fock vacuum annihilated by their adjoints. The Hilbert space is the exterior algebra $\mathcal{H} = \Lambda^\bullet V$ on the $3pn^2$ modes of $V = \mathbb{C}^p \otimes [(\mathbf{n}, \bar{\mathbf{n}}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{n}, \bar{\mathbf{n}}) \oplus (\bar{\mathbf{n}}, \mathbf{1}, \mathbf{n})]$, of dimension 2^{3pn^2} . The dynamics is fixed by a single supercharge,

$$Q = \sum_{a,b,c=1}^p C_{abc} \text{Tr}(A^a B^b C^c) = \sum_{a,b,c} C_{abc} A^a_{ij} B^b_{jl} C^c_{li}, \quad H = \{Q, Q^\dagger\} \geq 0. \quad (2)$$

Because Q is a product of three creation operators it acts on $\mathcal{H}$ as exterior multiplication by a fixed gauge-invariant three-form, and $Q^2 = 0$ holds identically: no condition is imposed on the couplings, which form an arbitrary tensor $C \in \mathbb{C}^p \otimes \mathbb{C}^p \otimes \mathbb{C}^p$. (The three fields are distinct, so no symmetrisation is implied, and flavour rotations $U(p)^3$ act on C .) All numerical results below use generic couplings, mostly random integers between 1 and 5 with a fixed seed, and several Gaussian real and complex sets as checks. Figure 1 shows the quiver and the interaction vertex.

The theory has a $U(1)^3$ of fermion numbers N_A, N_B, N_C , each raised by one unit by Q . Two combinations of them, $N_A - N_C$ and $N_B - N_A$, are the central charges of the gauge nodes, so gauge singlets have $N_A = N_B = N_C = m$ and total fermion number $k = N_A + N_B + N_C = 3m$. The total fermion number plays the role of the R-charge, and Q raises it by $\hat{q} = 3$. BPS states, $H\psi = 0$, are in one-to-one correspondence with Q -cohomology classes, and we write h^k for their number in degree k and $n(k)$ for the number of gauge singlets in degree k . The pairing with the top form of $\Lambda^\bullet V$ exchanges k and $3pn^2 - k$, so $n(k)$ is palindromic and $h^k = h^{3pn^2 - k}$, and half filling is $k_* = 3pn^2/2$. Non-BPS states come in Q -pairs $(\psi, Q\psi)$ in degrees k and $k+3$ with a common energy, and we label the pair by its charge relative to half filling, $q = k + \frac{3}{2} - \frac{3}{2}pn^2$, following [4]. Since $(-1)^k$ anticommutes with Q , the singlet index

$$I_0(n, p) = \sum_m (-1)^m n(3m) = \sum_m (-1)^m h_{\text{sing}}^{3m} \quad (3)$$

is a lower bound on the number of singlet BPS states. The bound is saturated when the singlet BPS states sit in a single degree, which is what concentration means here. In $\mathcal{N} = 2$ SYK with $\hat{q} = 3$ the BPS states fill three adjacent charges in the ratio 1:2:1 [3, 4]; gauge singlets live only in degrees $k \equiv 0 \pmod{3}$, so for even pn^2 that window contains exactly one singlet degree, half filling, and concentration of the singlet sector means that every

singlet BPS state has $k = 3pn^2/2$. By Molien–Weyl, $n(t) = \sum_k n(k)t^k$ is the unitary matrix integral

$$n(t) = \int \prod_{v=0}^2 dU_v \prod_{v=0}^2 \det(1 + t U_v \otimes U_{v+1}^\dagger)^p, \quad I_0 = n(-1), \quad (4)$$

over Haar measure, with v taken mod 3. This integral is the starting point of Section 6.

Two structural remarks explain why the model was chosen and where its interesting range of parameters lies. The weights of the bifundamental modes are differences $e_i - f_j$ of weights of two different nodes and never vanish, so the zero-weight (Cartan) modes on which the window-widening mechanism of the adjoint models lives are simply absent. This removes the obstruction of [6, 7]; it does not by itself imply concentration. Second, the gauge group has dimension $3n^2$ against $3pn^2$ fermion modes, a ratio of p rather than a large number, so the singlet constraint is severe. At $p = 1$ it leaves essentially nothing; at $n = 2$ and 3 every irrep’s complex has index ± 1 , the singlet index vanishes, and concentration is vacuous. The model becomes interesting at $p \geq 2$.

3 Relation to previous work

Witten obtained the large- N physics of SYK without disorder from a coloured model of $q = D+1$ fermion fields of rank D , every pair of which shares one index [8]. For $q = 3$ and $D = 2$ the three fields are exactly bifundamentals of a three-node quiver and the vertex is $\text{Tr}(ABC)$. Witten’s melonic dominance argument, however, requires $D \geq 3$ (his eq. (3.10)); at $D = 2$ the relevant combinatorial invariant reduces to the genus of ribbon graphs and the large- n limit is planar. Our model is the complex, flavoured, $\mathcal{N} = 2$ supersymmetric version of this matrix member of the coloured family. Supersymmetric tensor versions of SYK exist with $\mathcal{N} = 1$ supersymmetry and higher-rank tensors [9]. Chang, Colin-Ellerin and Rangamani showed that $\mathcal{N} = 2$ supertensor models with dynamical bosons break supersymmetry at large N , and noted that the purely fermionic $\mathcal{N} = 2$ model with an odd-degree supercharge has no evident melonic tensor version (their eq. (6.4)) [10]. Biggs, Lin and Maldacena have since closed that gap melodically, replacing tensor contractions by $SU(2)$ $3j$ symbols [11]. The quiver is a planar counterpart: purely fermionic, cubic and $\mathcal{N} = 2$, without dynamical bosons and hence without the supersymmetry-breaking mechanism of [10].

The same three-node quiver with a closed loop and a cyclic cubic superpotential is famous in a different guise, as the quiver quantum mechanics of three-centre BPS bound states in string theory. For bosonic chiral multiplets on abelian nodes, Bena, Berkooz, de Boer, El-Showk and Van den Bleeken found exponentially many “pure-Higgs” states that are invisible on the Coulomb branch and are natural candidates for single-centre black hole microstates [12]; they also remarked that their counting formula suggests an origin in fermionic degrees of freedom. Our model is a different theory, purely fermionic and non-abelian, whose BPS states are the cohomology of a finite Fock space with no Higgs or Coulomb branch, and we claim no equivalence; a naive comparison of our $n = 1$ counts with theirs already fails.

On the fortuity side, the monotone/fortuitous dichotomy and the covering construction are due to Chang and Lin [1], and concentration as its signature in SYK to [2]. Chen constructed fortuitous states in a single adjoint matrix and used a unitary matrix integral to follow them as the rank changes [5]. Tierz computed BPS spectra of $\text{Tr}\Psi^p$ models and formulated fortuity through projections between ranks, which are cochain maps [13]; we use his definition. Choi, Choi and Kim studied fortuity in the bifundamental conifold theory and introduced “generalised monotones” dressed by baryons [14], which bears directly on our fortuity test (Section 4.3). The natural tool for large n is the quantum mechanical matrix bootstrap [15, 16, 17], which Chen already suggested for matrix SYK models; our sector-resolved version is described in [7]. A literature search in October 2026 found no previous study of the fermionic quiver itself, although absence from searches is not proof.

4 BPS states at small rank

4.1 Singlet indices

The singlet multiplicities $n(k)$ follow from the dual Cauchy identity, $\Lambda(\mathbb{C}^p \otimes V_v \otimes \bar{V}_{v+1}) = \bigotimes_{f=1}^p \bigoplus_\lambda S_\lambda V_v \otimes S_{\lambda'} \bar{V}_{v+1}$, which reduces the singlet count to products of Littlewood–Richardson coefficients. We compute these by exact quadrature on the torus and assemble the index with a transfer matrix over $U(n)$ irreps that adds one flavour at a time,

$$I_0 = \text{Tr}[(R^{(p)})^3], \quad R^{(q)} = \sum_\lambda (-1)^{|\lambda|} L_\lambda^T R^{(q-1)} L_{\lambda^T}, \quad (L_\lambda)_{\mu\nu} = c_{\mu\lambda}^\nu, \quad (5)$$

(n, p)	modes $3pn^2$	singlet states	$I_0(n, p)$	$\ln I_0 $ per mode	$(-1)^{pn^2/2}$
(1, 2)	6	10	-6	0.299	-
(2, 2)	24	1 274	90	0.187	+
(3, 2)	54	2 115 632	-1 680	0.138	-
(4, 2)	96	5.59×10^{10}	34 650	0.109	+
(1, 3)	9	56	0	-	$(pn^2 \text{ odd})$
(2, 3)	36	889 040	1 680	0.206	+
(3, 3)	81	4.90×10^{12}	0	-	$(pn^2 \text{ odd})$
(4, 3)	144	1.08×10^{22}	9 434 197 872	0.159	+
(1, 4)	12	346	90	0.375	+
(2, 4)	48	9.70×10^8	519 750	0.274	+
(3, 4)	108	2.99×10^{19}	199 090 719 360	0.241	+
(1, 5)	15	2 252	0	-	$(pn^2 \text{ odd})$
(2, 5)	60	1.37×10^{12}	95 866 056	0.306	+
(1, 6)	18	15 184	-1 680	0.413	-
(2, 6)	72	2.29×10^{15}	28 686 212 100	0.334	+

Table 1: Exact gauge-singlet indices and singlet counts. The $n = 1$ rows are Dixon’s sum $\sum_m (-1)^m \binom{p}{m}^3$ and the Franel numbers $\sum_m \binom{p}{m}^3$. At $p = 1$ the index vanishes for $n = 2, 3, 4$. The last column is the sign that a singlet BPS sector concentrated at half filling would give; it agrees with every non-zero entry.

in exact integer arithmetic. This reaches $(n, p) = (4, 3)$ and $(3, 4)$ and reproduces every value obtained earlier by direct weight counting. Table 1 lists the results together with the total singlet counts.

Three features of the table are worth stating at once. First, the $p = 2$ entries are $(-1)^n (3n)!/(n!)^3$, a formula that we found numerically and prove in Section 6. The first few indices at $p = 3$ coincided with this pattern, $|I_0(2, 3)| = |I_0(3, 2)| = 1680$, which briefly suggested that the index depends only on np and grows only exponentially in n ; the value at $(4, 3)$, which is about 550 times larger than the $p = 2$ -type value, shows that the coincidence was accidental. Second, whenever pn^2 is even the sign of I_0 is $(-1)^{pn^2/2}$, which is what a singlet BPS sector concentrated at half filling would produce, since half filling has $m = pn^2/2$; for odd pn^2 the index vanishes identically, because the particle-hole map $m \rightarrow pn^2 - m$ pairs degrees of opposite sign. Third, at fixed n the index per mode rises with p towards the singlet entropy per mode, which already suggests that large p is the SYK-like direction.

4.2 Concentration at $(2, 2)$ and $(2, 3)$

At $(n, p) = (2, 2)$ the singlet cohomology can be computed with exact (modular) ranks. There are exactly 90 singlet BPS states, all at $k = 12$, which is half filling, so the index is saturated. At $(2, 3)$ the Fock space has 2^{36} states and the singlet sector at half filling has 300 584, which calls for a different method. We restrict to the zero-weight subspace of each degree, which contains every irrep, and to the vectors invariant under the Weyl group $(S_2)^3$, which commutes with Q and fixes singlets; this reduces the dimension by about a factor of eight. On these vectors the operator

$$P = \sum_{v=0}^2 \|E_{12}^{(v)} \psi\|^2 = \sum_v \langle \psi | J_v^2 | \psi \rangle \quad (6)$$

is the sum of the $SU(2)$ Casimirs of the three nodes, because $J_- J_+ = J^2$ at $J_z = 0$. It commutes with H , vanishes exactly on singlets and is at least 2 on every non-singlet. The kernel of $K = H + \mu P$ is therefore the singlet BPS space, and every eigenvalue of K below 2μ is an exact singlet energy. Lanczos diagonalisation of K in the degrees $k = 9, 12$ and 15 (Weyl-symmetric dimensions 5.6×10^4 , 5.1×10^5 and 1.8×10^6), and dense diagonalisation for $k \leq 6$, finds no zero modes, with lowest eigenvalues 96.5, 0.3348 and 0.05386 at $k = 9, 12$ and 15 ; reruns at $k = 12$ and 15 with different starting vectors reproduce them to nine digits. Particle-hole symmetry excludes BPS states at $k \geq 21$ as well, so all singlet BPS states sit at $k = 18$, and by the index there are exactly $h_{\text{sing}}^{18} = |I_0| = 1680$ of them. Figure 2 summarises both cases: the singlet sector spreads over 13 degrees at $(2, 3)$, but its BPS states occupy one.

Zero-weight methods stop at $n = 2$. At $(3, 2)$ the zero-weight sectors near half filling hold 5×10^{10} states, although the singlet subspaces are only about 5×10^5 , so what is needed is an explicit basis of singlets. Section 7

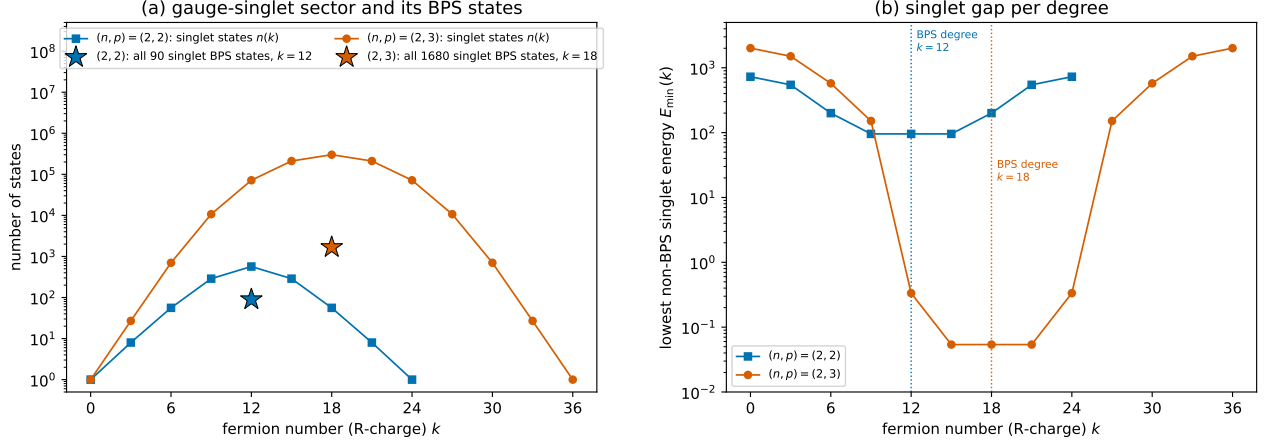


Figure 2: Concentration of the singlet BPS states at $n = 2$. (a) The number of gauge singlets in each degree k (lines) and the singlet BPS states (stars): 90 at $k = 12$ for $p = 2$ (exact ranks) and 1680 at $k = 18$ for $p = 3$ (Lanczos); there are none in any other degree. (b) The lowest non-BPS singlet energy in each degree. At $p = 3$ the gap collapses by four orders of magnitude next to the BPS degree, to 0.054 at $k = 15, 18, 21$ and 0.335 at $k = 12, 24$, while at $p = 2$ it never falls below 95.6.

describes one, and the bootstrap that takes over where even the singlet spaces are too large.

4.3 Fortuity at $(2, 2)$

Following Tierz [13], we call a rank-2 class monotone only if it lies in the image of the projection $\pi_{3 \rightarrow 2}$ that deletes every monomial involving a third index on any node. Because Q contains only creation operators, $\pi Q^{(3)} = Q^{(2)}\pi$, so the projection is a cochain map. Writing $Q^{(3)} = Q^{(2)} + Q_{\text{new}}$, a closed rank-2 state z_0 is the projection of a closed rank-3 state if and only if $Q_{\text{new}}z_0$ is exact in the subspace of states that carry index-3 modes. Grading that subspace by the number of index-3 modes, the first condition arises with two such modes, and for couplings of full rank in the summed flavour index it reduces to the vanishing of the BPS three-point function $\langle \beta | X | z_0 \rangle$ for every single letter X and every BPS state β in degree $k + 1$. A non-vanishing value obstructs the lift. By gauge covariance it suffices to test one component of each letter. At $(2, 2)$, each of the six letters obstructs a 54-dimensional subspace of the 90 classes, different letters obstruct different subspaces, and together they obstruct all 90. Since the obstruction appears already at the first order of the lift, none of the 90 classes lies in the image of $\pi_{3 \rightarrow 2}$: every singlet BPS class at $(2, 2)$ is fortuitous.

This result is expected rather than discriminating, and the reason can be made precise. The projections preserve the degree, while the half-filling degree $k_* = 3pn^2/2$ moves by $3p(2n+1)/2$ when the rank grows by one, more than the width of any concentrated window. A rank- n class of degree $k_*(n)$ can only be monotone if there is a class of the same degree at rank $n+1$, so concentration at rank $n+1$ implies that every singlet BPS class at rank n is fortuitous; in this sense fortuity follows from concentration, much as in Chang–Chen–Sia–Yang’s argument for generic SYK [2]. At $(2, 2) \rightarrow (3, 2)$ the relevant rank-3 singlet space, at $k = 12$, has only 3494 states, and Section 7 checks the statement directly. The $(2, 2)$ test itself uses one coupling and floating-point ranks, and it is in the ordinary sense. Under the refined criterion of [14], classes of the form (determinant ground state) $\times$ (n -independent excitation) would count as generalised monotones; with fermionic letters many determinants vanish, and whether any of the 90 classes is of that form has not been analysed.

5 The singlet spectrum: chaos at $p = 3$, hidden supersymmetry at $p = 2$

The exactness of the Casimir penalty makes the full singlet spectrum accessible at $n = 2$. Where the singlet subspace is small we construct it explicitly as $\ker P$ and diagonalise $(QU_k)^T(QU_k)$ on a singlet basis U_k , whose non-zero eigenvalues are the multiplet energies of the pair $(k, k+3)$. For the largest pair at $(2, 3)$, $(9, 12)$ with 10700 singlets in the lower degree, we project random vectors onto singlets with the exact polynomial $\prod_v \prod_{j=1}^6 (1 - J_v^2/j(j+1))$, which leaves a non-singlet residual of 1.4×10^{-15} , and compress $Q^T Q$ onto their span.

Every count checks: the rank of the projected vectors equals $n(9)$, the kernel equals the rank of the previous map, and the lowest level, 151.34, agrees with Lanczos.

At $p = 3$ the multiplet spectra are chaotic. The mean ratio of consecutive level spacings over the central 80% of the 10 024 multiplets of the pair (9, 12) is $\langle r \rangle = 0.5268 \pm 0.0028$. The Gaussian-orthogonal value for matrices of the same size is 0.5343 ± 0.0026 and the Poisson value 0.3869 ± 0.0042 , so the spectrum is about 30 standard deviations from Poisson and within two of the orthogonal class. The smaller pair (6, 9), with 676 levels, gives 0.507 ± 0.011 , and the locally unfolded spacing distribution follows the Wigner surmise (Figure 3c). The orthogonal class is the expected one, since for real couplings H is a real matrix in the occupation-number basis.

The near-BPS structure is equally clear. The lowest multiplet energies of the pairs at (2, 3) are 0.05386 for $|q| = 1.5$, the pairs adjacent to the BPS degree, then 0.3348 for $|q| = 4.5$, and then a jump to the microscopic scale set by the couplings: 151.34, 575.23, 1506 and 2008 for $|q| = 7.5, 10.5, 13.5$ and 16.5. Small gaps exist, and only next to the BPS degree (Figure 2b). A quantitative test of the Turiaci–Witten density [4], $\rho_q(E) \propto \sinh(2\pi\sqrt{(E - E_0(q))/E_s})/E$ with $E_0(q) = q^2 E_s/4\hat{q}^2$, does not succeed at this size. Fixing $E_s = 0.862$ from the $|q| = 1.5$ edge predicts $E_0 = 0.485$ at $|q| = 4.5$, against 0.335 measured, an edge ratio of 6.2 instead of 9; fixing the normalisation from the eleven lowest $|q| = 1.5$ levels predicts 5.4, 258, 2000 and 8.7×10^4 multiplets with $|q| = 4.5$ below $E = 1.0, 2.17, 3.11$ and 5.38, against 2, 3, 5 and 7 measured (Figure 3d). This is not surprising. The Turiaci–Witten density is a statement at large e^{S_0} in which neighbouring charge sectors carry the same entropy, whereas at (2, 3) the singlet sectors at $k = 12, 15$ and 18 hold 71 685, 211 113 and 300 584 states, changing by factors of three to four per step. The test needs larger n .

The case $p = 2$ behaves very differently, and the reason is an unexpected symmetry. The (2, 2) spectra are full of exact coincidences between different pairs, and a search over cubic supercharges $\tilde{Q} = Q(\tilde{C})$ that commute with H on singlets finds exactly one besides Q , namely $\tilde{C} = (\varepsilon \otimes \varepsilon \otimes \varepsilon)\tilde{C}$ with ε the 2×2 antisymmetric tensor. On every singlet sector, to 10^{-15} , and for integer, Gaussian real and Gaussian complex couplings,

$$\tilde{Q}^2 = 0, \quad \{Q, \tilde{Q}\} = 0, \quad \{Q^\dagger, \tilde{Q}\} = 0, \quad \{\tilde{Q}, \tilde{Q}^\dagger\} = H, \quad (7)$$

which is an $\mathcal{N} = 4$ algebra. Off the singlets the last two relations fail by about 30% of $|H|$, so the extended supersymmetry is a property of the gauged model, closing only up to gauge transformations. The singlet spectrum organises completely into long $\mathcal{N} = 4$ multiplets spanning degrees $k, k + 3, k + 3$ and $k + 6$, with $n(k) = b(k) + 2b(k - 3) + b(k - 6) + h(k)$ in every degree. Once the flavour symmetry $F = \varepsilon$ on every edge (or the corresponding antiunitary symmetry for complex couplings) is resolved, the level statistics are still not random-matrix like in any symmetry class, with coupling-independent exact degeneracies, and within the span of all gauge-invariant operators with at most four fermions the commutant of H on the $k = 9$ singlets is just $\{1, H\}$. At $p = 3$ the same search finds only Q . The hidden supersymmetry is plausibly tied to the pseudoreality of the $SU(2)$ flavour doublet, but we have not derived it; it holds at $n = 2$ and, on every singlet degree below half filling, at $n = 3$ (Section 7).

6 Large n : the singlet index as a three-species log gas

6.1 An exact rewriting

Write $x_{vi} = e^{i\theta_{vi}}$ for the eigenvalues of U_v in (4) at $t = -1$. Weyl's integration formula introduces $|\Delta(x_v)|^2/n!$ for each node, and the edge factors become $\prod_{ij} (1 - e^{i(\theta_{vi} - \theta_{v+1,j})})^p$. For real φ , $1 - e^{i\varphi} = 2 \sin(\varphi/2) e^{i(\varphi - \pi)/2}$, and the φ -dependent phases cancel around the triangle, since each node is the source of one edge and the target of another. Using $|x_a - x_b| = |2 \sin \frac{\theta_a - \theta_b}{2}|$,

$$I_0(n, p) = \frac{(-i)^{3pn^2}}{(n!)^3} \int \prod_{v,i} \frac{d\theta_{vi}}{2\pi} e^{L(\theta)} \sigma(\theta)^p, \quad (8)$$

$$L = \sum_{\substack{\text{pairs at} \\ \text{the same node}}} 2 \ln |2 \sin \frac{\theta_a - \theta_b}{2}| + p \sum_{\substack{\text{pairs at} \\ \text{different nodes}}} \ln |2 \sin \frac{\theta_a - \theta_b}{2}|,$$

with $\sigma = \prod_v \prod_{ij} \text{sgn} \sin \frac{\theta_{vi} - \theta_{v+1,j}}{2}$. This is the partition function of a gas of $3n$ charges of three species on the circle: charges of the same species repel with weight 2 and charges of different species with weight p , and for odd p there is an additional sign. Because every pair of distinct nodes is joined by exactly one edge, e^L is symmetric under permutations of the species, and only σ remembers the orientation of the quiver. The singlet count is the same integral with $|2 \cos \frac{\theta_a - \theta_b}{2}|$ for pairs at different nodes, which attract, and no phase.

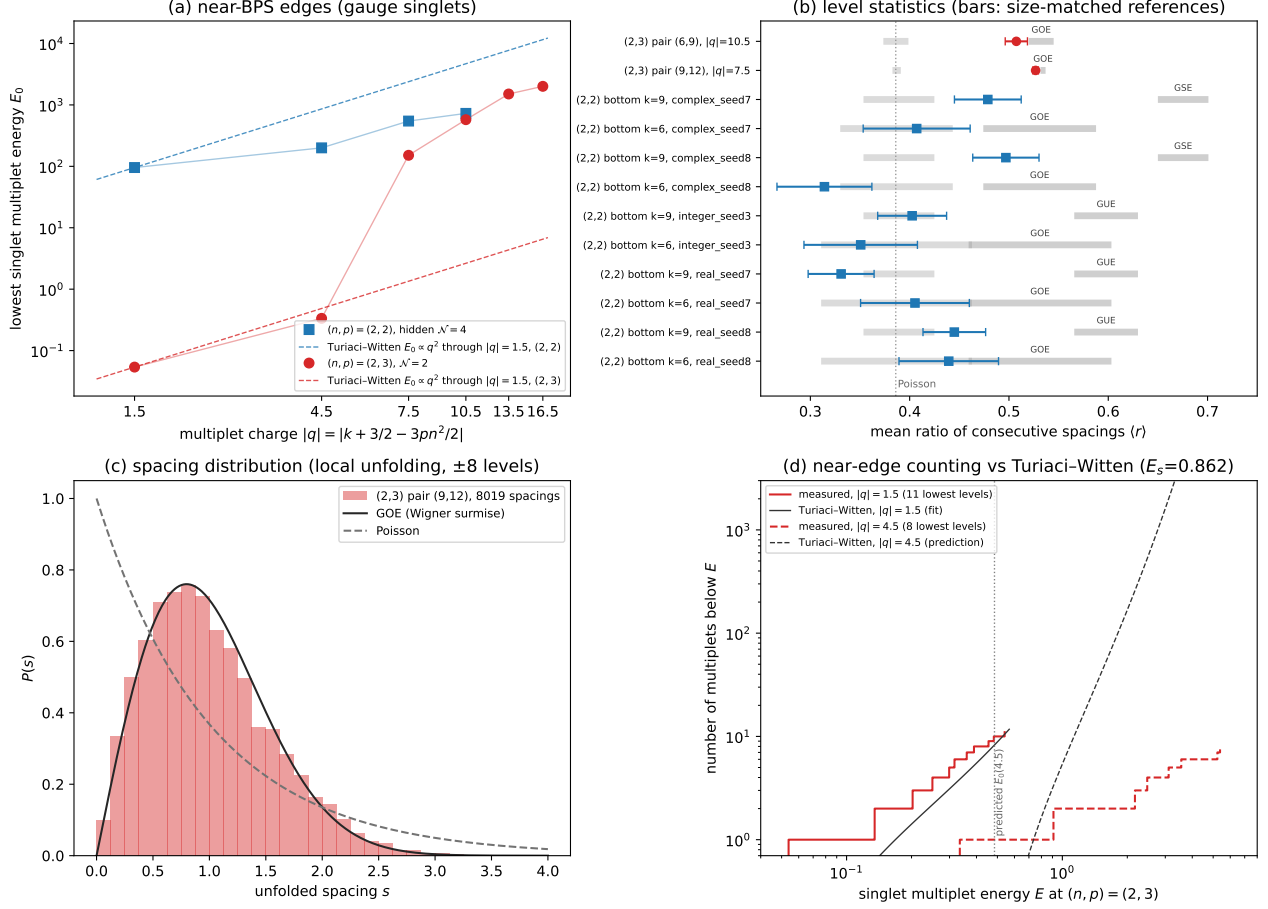


Figure 3: Singlet spectra at $n = 2$. (a) Lowest singlet multiplet energy of each Q -pair against its charge $|q|$, with the Turiaci–Witten q^2 law drawn through the $|q| = 1.5$ point. (b) Mean spacing ratios of every symmetry-resolved ensemble, with size-matched random-matrix and Poisson references. (c) Spacing distribution of the 10024 multiplets of the pair (9, 12) at (2, 3), locally unfolded, against the Wigner surmise and Poisson. (d) Counting functions of the two innermost multiplet sectors at (2, 3) against the Turiaci–Witten density, normalised on $|q| = 1.5$; the $|q| = 4.5$ curve is a prediction.

Several exact statements follow immediately. At $p = 2$ every pair carries weight 2 and $\sigma^2 = 1$, so e^L is the Vandermonde $|\Delta_{3n}(x)|^2$ of all $3n$ eigenvalues together. Its integral is $(3n)!$, and

$$I_0(n, 2) = (-1)^n \frac{(3n)!}{(n!)^3} \quad \text{for every } n. \quad (9)$$

As measured by the index, the singlet BPS entropy of the $p = 2$ model is therefore $3n \ln 3 + O(\ln n)$, which is not macroscopic. The same manipulation, using $(1 - x\bar{y})^2 = -x\bar{y}|x - y|^2$ on the unit circle, gives for general p

$$I_0(n, p) = (-1)^n \frac{(3n)!}{(n!)^3} \mathbb{E}_{\text{CUE}(3n)} \left[\prod_v \prod_{i,j} (1 - x_{vi} \bar{x}_{v+1,j})^{p-2} \right], \quad (10)$$

where the $3n$ eigenvalues of a Haar-random $U(3n)$ matrix are split into three labelled groups. For even p the integrand of (8) is positive, so $I_0 \neq 0$ with sign $(-i)^{3pn^2} = (-1)^{pn^2/2}$; for odd p and odd n the prefactor is $\pm i$ while the integral is real, so $I_0 = 0$; and for odd p , Cauchy–Schwarz in p gives $|I_0(n, p)|^2 \leq |I_0(n, p-1)| |I_0(n, p+1)|$, which holds in the data with ratios 16.6 at (2, 3) and 1.62 at (2, 5). These are exactly the sign rules observed in Table 1.

p	F^* (index)	arc half-width	G^* (singlets)	F^*/G^*	$F^*/3p$	$G^*/3p$
2	0 (exact)	$\pi/3$	1.4244	0	0	0.237
3	1.2373	1.0276	3.0427	0.41	0.137	0.338
4	2.5550	0.9717	4.7688	0.54	0.213	0.397
5	3.9297	0.9132	6.5623	0.60	0.262	0.437
6	5.3455	0.8606	8.4015	0.64	0.297	0.467
8	8.2619	0.7747	12.1714	0.68	0.344	0.507
16	20.4849	0.5796	27.8331	0.74	0.427	0.580
∞	$\frac{3p}{2} \ln 3 - \frac{3}{2} \ln \frac{2p}{3} - \frac{9}{4}$	$\sqrt{6/p}$	$3p \ln 2 - \frac{3}{2} \ln \frac{p}{2} - \frac{9}{4}$	$\frac{\ln 3}{2 \ln 2}$	$\frac{1}{2} \ln 3$	$\ln 2$

Table 2: Leading large- n coefficients, $\ln |I_0| = F^* n^2$ and $\ln(\#\text{singlets}) = G^* n^2$ up to $o(n^2)$, from the continuum solution, with the large- p asymptotics in the last row. The last two columns are per fermion mode.

6.2 The saddle point

At large n , with normalised eigenvalue densities ρ_v , $L = n^2 F[\rho] + O(n \ln n)$ with

$$F[\rho] = \sum_v \langle \rho_v, f \rho_v \rangle + p \sum_{v < w} \langle \rho_v, f \rho_w \rangle, \quad f(\alpha) = \ln \left| 2 \sin \frac{\alpha}{2} \right| = - \sum_{m \geq 1} \frac{\cos m\alpha}{m}, \quad (11)$$

and since $(n!)^3 = e^{O(n \ln n)}$, $\ln |I_0| = n^2 \max F + o(n^2)$ for even p by standard log-gas large deviations. In terms of the moments $u_{v,m} = \int \rho_v e^{im\theta}$, $F = - \sum_m \frac{1}{m} u_m^\dagger [\mathbb{K} + \frac{p}{2}(J - \mathbb{K})] u_m$ with J the all-ones 3×3 matrix, whose eigenvalues are $1+p$ for the species-symmetric combination and $1-p/2$ twice. For $p < 2$ the uniform distribution is a stable maximum with $F = 0$; at $p = 2$ it is marginal; for $p > 2$ it is unstable towards configurations in which the species separate, in the spirit of the Hagedorn instability of weakly coupled gauge theories [18]. The maximum then sits on the boundary of the positivity constraint: the eigenvalues of the three nodes occupy three disjoint arcs rotated by $2\pi/3$ with respect to each other (Figure 4d). Physically the holonomies sit near $U_v \approx e^{2\pi i v/3}$, which turns the factor (-1) that each bifundamental fermion carries in the index into a phase $e^{\mp 2\pi i/3}$ and so evades the cancellations between even and odd states that keep the index small at $p \leq 2$.

We solve the continuum problem by expanding the density of one arc in Chebyshev polynomials. The Cauchy part of the Euler–Lagrange equation is treated exactly and the smooth part by quadrature; the arc width follows from the condition that the edges be soft, and F^* is three times the Lagrange multiplier. The solution converges to twelve digits, and the effective potential is flat on the support to 10^{-15} and lower in the gaps. Independently, Newton maximisation of the discrete L up to $n = 2048$ extrapolates to the same coefficients to 10^{-9} or better, and for $n \leq 12$ no unconstrained random start finds a higher maximum than the symmetric one. The singlet count is treated in the same way: there the three species coincide on a single arc. Table 2 gives the coefficients.

Both limits can be derived analytically and are reproduced by the numerics. Near $p = 2$ the uniform arcs of length $2\pi/3$ give the bound $F^*(p) \geq (p/2 - 1) 39\zeta(3)/2\pi^2$, which is tight as $p \rightarrow 2^+$. At large p the arcs shrink to width $O(p^{-1/2})$, each becomes a semicircle of radius $\sqrt{6/p}$ in the effective harmonic potential of the other two, and $F^* = \frac{3p}{2} \ln 3 - \frac{3}{2} \ln \frac{2p}{3} - \frac{9}{4} + O(1/p)$; the singlet count behaves in the same way with $\ln 2$ in place of $\frac{1}{2} \ln 3$. The leading terms have a simple meaning. $3p \ln 2 n^2$ is the logarithm of the dimension of the whole Fock space, and $\frac{3p}{2} \ln 3 n^2$ is what $3pn^2$ fermion modes contribute when each carries the weight $|1 - e^{-2\pi i/3}| = \sqrt{3}$. The BPS entropy per fermion therefore approaches $\ln(2 \cos \frac{\pi}{6}) = \frac{1}{2} \ln 3$, which is precisely the ground-state entropy per complex fermion of $\mathcal{N} = 2$ SYK with $\hat{q} = 3$ [3], and the ratio F^*/G^* approaches the SYK ratio $\ln \sqrt{3}/\ln 2 = 0.79$. The same constant appears in Tierz’s large- N result for the irrep-summed indices of the single-matrix $\text{Tr} \Psi^3$ model [13]. At finite p the gauge constraint costs $\frac{1}{2p} \ln \frac{2p}{3} + \frac{3}{4p}$ per mode, which is large at $p = 3$: there the singlet BPS entropy per mode is 0.137, a quarter of the SYK value, but it is a finite fraction, 41%, of the singlet log-count.

6.3 Finite n and the test at $(4, 3)$

The coefficient F^* alone cannot be compared with the data at $n \leq 4$: at $(4, 3)$ it gives $F^* n^2 = 19.80$ against the exact $\ln |I_0| = 22.97$, a factor of 24 too small. The subleading terms come from the same saddle. The discrete maximum behaves as $L_{\max} = F^* n^2 + 3n \ln n + 3n \int g \ln(2\pi g) + \frac{1}{4} \ln n + O(1)$, where g is the density of one arc: the $n \ln n$ and $O(n)$ terms are the self-energy of each species at local spacing $1/(ng)$, which we verified to six digits, and the $\frac{1}{4}$ is consistent with $\frac{1}{24}$ per soft edge. Adding the Gaussian fluctuations about the maximum overshoots by a universal amount, the excess of the harmonic approximation for a $\beta = 2$ log gas, which on the circle is $\ln(e/\sqrt{2\pi})$ per eigenvalue up to $O(1/n)$. With that correction the estimate reproduces (9) to $1/36n$, and

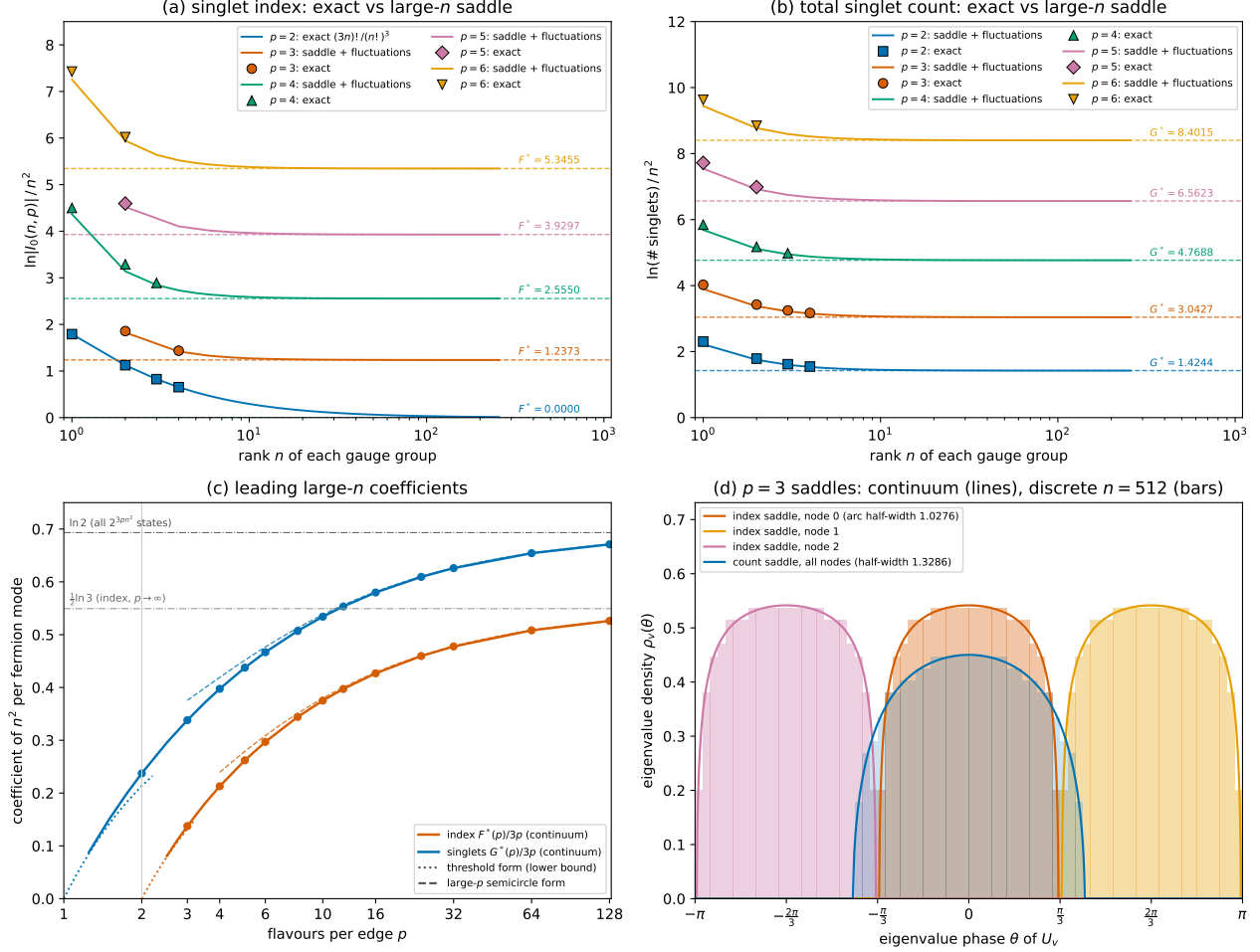


Figure 4: The large- n saddle. (a) $\ln|I_0|/n^2$ against n : exact values (markers), the saddle-point estimate with Gaussian fluctuations and the universal crystal correction (lines), and the asymptotes $F^*(p)$ (dashed); for $p=2$ the line is the exact result (9). (b) The same for the total singlet count. (c) The leading coefficients per fermion mode against p , with the threshold bound and the large- p asymptotics. (d) The equilibrium densities at $p=3$: the three index arcs and the single arc of the count saddle, with histograms of the discrete maximum at $n=512$.

for $p \neq 2$ we assume that each arc is locally a $\beta=2$ gas, which is the one heuristic step. The resulting estimate has no free parameters, and Table 3 compares it with every non-zero exact value we have. All 27 residuals are negative and none exceeds 0.6 in the logarithm, across values from $e^{1.8}$ to $e^{50.7}$; at $(4, 3)$ the estimate gives 22.69, within 25% of $|I_0| = 9.43 \times 10^9$. At large n the $O(n)$ terms cancel to fit precision and the estimate takes the form $\ln|I_0| \approx F^* n^2 + \frac{1}{2} \ln n + d(p)$, with $d \approx 2.2$ – 2.4 for $p=3$ – 6 , in line with the absence of an $O(n)$ term in $\beta=2$ ensembles. For $p=3$ it predicts $\ln|I_0| \approx 47.7, 82.6, 127.3$ and 181.8 at $n=6, 8, 10$ and 12 , beyond the reach of the exact recursion.

The status of these statements should be clear. The rewriting (8), the formula (9) and the sign rules are exact. The coefficient F^* for even p rests on standard large-deviation results for log gases, which we have not reproved. For odd p it also requires that the sign σ^p does not cancel the leading saddle. Sign changes need an eigenvalue of one node to pass an eigenvalue of a neighbouring node across the gap between arcs, which costs an action of order n ; but the gap at $p=3$ is only 0.039 radians, so the suppression is modest at the sizes of interest. The exact values at $(2, 3)$ and $(4, 3)$ show no sign of cancellation, their residuals being like those at even p . Finally, $|I_0|$ is a lower bound on the number of singlet BPS states, equal to it only in the absence of cancellations between degrees, for example under concentration; and for odd pn^2 the index vanishes and says nothing about that number.

(n, p)	$\ln I_0 $			$\ln(\#\text{singlets})$		
	exact	saddle	difference	exact	saddle	difference
(1, 2)	1.79	1.76	-0.03	2.30	2.22	-0.08
(2, 2)	4.50	4.49	-0.01	7.15	7.00	-0.15
(3, 2)	7.43	7.42	-0.01	14.56	14.40	-0.17
(4, 2)	10.45	10.45	-0.01	24.75	24.55	-0.19
(1, 3)		both vanish		4.03	3.90	-0.13
(2, 3)	7.43	7.30	-0.12	13.70	13.50	-0.20
(3, 3)		both vanish		29.22	28.97	-0.25
(4, 3)	22.97	22.69	-0.28	50.74	50.44	-0.30
(1, 4)	4.50	4.37	-0.13	5.85	5.69	-0.16
(2, 4)	13.16	12.56	-0.60	20.69	20.45	-0.24
(3, 4)	26.02	25.62	-0.39	44.85	44.54	-0.30
(1, 5)		both vanish		7.72	7.54	-0.18
(2, 5)	18.38	18.08	-0.29	27.94	27.68	-0.26
(1, 6)	7.43	7.26	-0.17	9.63	9.44	-0.19
(2, 6)	24.08	23.78	-0.30	35.37	35.09	-0.28

Table 3: The saddle-point estimate (Gaussian fluctuations about the discrete saddle, minus the universal $\beta = 2$ crystal excess, no free parameters) against all exact values. For odd p and odd n the index and the estimate both vanish exactly, the latter because the two orientations of the arcs cancel.

7 Beyond $n = 2$: a Schur–Weyl basis and a singlet bootstrap

7.1 Exact methods at rank three

Two tools take us past $n = 2$. The first is a singlet-adapted basis. By Schur–Weyl duality a singlet of degree $3m$ is a triple of elements of the group algebra of S_m , one for each node, recording how the m letters leaving the node are contracted with the m letters arriving at it. In Young’s orthogonal form the fermionic relabellings become sign projections on each edge, the finite- n trace relations become the condition that every Young diagram has at most n rows, and Q becomes a Kronecker product of Gelfand–Tsetlin inclusions, which is applied without ever building Fock vectors. The basis reproduces every singlet count and every stored $n = 2$ spectrum. At $(3, 2)$ Lanczos diagonalisation in this basis finds no singlet BPS states in any degree $k \leq 24$, so all $|I_0| = 1680$ of them sit at half filling, $k = 27$: in the $p = 2$ corner concentration survives the step to the next rank. Since the rank-3 cohomology vanishes at $k = 12$, the argument of Section 4.3 makes all 90 classes at $(2, 2)$ fortuitous to all orders, and the hidden $\mathcal{N} = 4$ of Section 5 holds at $n = 3$ on every singlet degree below half filling. At $p = 3$, however, the basis reaches $(3, 3)$ only for $k \leq 12$, where it again finds no BPS singlets (Table 5), because the singlet dimension grows to 1.2×10^8 at $k = 18$ and 10^{12} at half filling. Nor is there an algebraic shortcut. Upper semicontinuity bounds the generic cohomology by that of any special coupling, but the natural special coupling $C_{abc} = \delta_{ab}\delta_{bc}$, for which the singlet complex is the gauge-invariant part of a tensor power of the $p = 1$ complex, is not concentrated even at $(2, 3)$, where its singlet BPS states spread over the degrees $12 \leq k \leq 24$.

7.2 The semidefinite program

The second tool is a semidefinite bootstrap of the singlet sector, built on the trace-word algebra of [7]. Write $\Psi_{e,f}$ for the matrix of creation operators of edge e (the fields A, B, C for $e = 0, 1, 2$) and flavour f , whose row index lives at node e and column index at node $e + 1$, and $\bar{\Psi}_{e,f}$ for the matrix of adjoints, $(\bar{\Psi}_{e,f})_{yx} = (\Psi_{e,f})_{xy}^\dagger$. The only non-vanishing anticommutator contracts two indices of the same node, so products of these matrices are brought to a canonical form exactly as for the adjoint models, with a factor n for every closed index loop. Gauge-invariant operators are products of traces of closed walks on the triangle, and for a density matrix ρ supported on singlets of degree $k = 3m$ the expectation values $\phi(X) = \text{Tr} \rho X$ of these products are the unknowns. With real couplings every such operator is a real matrix in the occupation basis and every eigenspace of H has a real basis, so ϕ may be taken real, which halves the problem relative to the complex engine of [7].

Positivity enters through Gram matrices. For open words W_a , products of letters that form a walk from node u to node v , and for traced closed walks T_a together with the identity,

$$M_{ab} = \phi(\text{Tr} W_a^\dagger W_b) - \frac{\delta_{uv}}{n} \phi(\text{Tr} W_a^\dagger \text{Tr} W_b) \succeq 0, \quad S_{ab} = \phi(T_a^\dagger T_b) \succeq 0. \quad (12)$$

(n, p)	k	exact E_0	bound on E_0	margin t^*	verdict
(2, 2)	3	546	455.1	-1.334	excluded
	6	199.95	182.6	-0.508	excluded
	9	95.56	0	0	not excluded
	12	0 (90 BPS states)	0	0	BPS degree, kept
(2, 3)	3	1506	1129.3	-1.342	excluded
	6	575.23	253.7	-0.732	excluded
	9	151.34	14.4	-0.163	excluded
	12	0.3348	0	0	not excluded
	15	0.0539	0	0	not excluded
	18	0 (1680 BPS states)	0	0	BPS degree, kept

Table 4: Validation of the level-2 singlet bootstrap against the exact singlet spectra at $n = 2$, for the seed-3 integer couplings used throughout. The bound is the minimum of $\phi(H)$ without the BPS conditions; the margin is computed with them, and $t^* = 0$ stands for $|t^*| < 10^{-6}$. Degrees above half filling follow by particle-hole symmetry.

M is the Gram matrix of the vectors $(W_a)_{ij}\psi$ summed over the open indices, with the trace part removed when the word returns to its starting node. Words with different endpoints cannot be combined, and words with different edge charges have orthogonal images because ρ has definite N_A , N_B and N_C , so both matrices are block diagonal in these labels. The linear constraints are the normalisation $\phi(1) = 1$; reality; the sector conditions $\phi((N_e - m)X) = 0$ for the three edge numbers; the gauge conditions $\phi(\hat{C}_v X) = 0$, where $\hat{C}_v = n \text{Tr} X_v^2 - (\text{Tr} X_v)^2$ is n times the $SU(n)$ Casimir of node v , built from its generator matrix $X_v = \sum_f (\Psi_{v,f} \bar{\Psi}_{v,f} + \bar{\Psi}_{v-1,f} \Psi_{v-1,f})$; and the equations of motion $\phi([H, X]) = 0$, which hold for any mixture of eigenstates. Minimising $\phi(H)$ subject to all of these gives a lower bound on the lowest singlet energy $E_0(k)$. For exclusion we add the BPS conditions

$$\phi(XQ) = \phi(QX) = \phi(XQ^\dagger) = \phi(Q^\dagger X) = \phi(XH) = \phi(HX) = 0, \quad (13)$$

which every BPS state satisfies. A singlet BPS state in degree k would then supply a feasible ϕ , so infeasibility proves $h_{\text{sing}}^k = 0$. We test it in two ways: through the margin t^* , the largest t for which every Gram matrix can be made $\succeq t\mathbf{1}$, with $t^* < 0$ meaning infeasible; and through the pure feasibility problem, whose infeasibility comes with a Farkas certificate. Writing the problem as $Ax + s = b$ with s in the product K of a zero cone and the semidefinite cones, the certificate is a vector y with $A^T y = 0$, $y \in K^*$ and $b \cdot y < 0$. Exclusion at one coupling suffices: $h^k = \dim V_k - \text{rk } Q_k - \text{rk } Q_{k-3}$ is upper semicontinuous in C , so its vanishing at our integer couplings implies its vanishing for generic couplings.

All results below are at what we call level 2: open words of length at most two, closed walks of length at most three, and constraints generated by operators of length at most four. At $p = 3$ the largest Gram blocks then have 37, 27 and 18 rows, and the exclusion problem has 1.6×10^5 unknowns and 1.2×10^5 equality constraints after the reality reduction. The interior-point solver Clarabel solves it in one to nine minutes at a peak of about 3 GB. Every family of constraints has been evaluated on exact functionals, computed from Fock-space states at $(n, p) = (1, 2)$, $(1, 3)$ and $(2, 1)$, and holds to 10^{-15} .

7.3 Validation at $n = 2$

Table 4 compares the bootstrap with the exact singlet spectra at $n = 2$. There are no contradictions: no degree that carries BPS states is excluded, and every energy bound lies below the exact gap, by 17% and 25% at $k = 3$ and by more towards the frontier. Level 2 excludes $k \leq 6$ at $(2, 2)$ and $k \leq 9$ at $(2, 3)$, in both cases half of the way to half filling. At $(2, 3)$ it stops precisely where the exact gap collapses from 151 to 0.33, but at $(2, 2)$ it also stops at $k = 9$, where the gap is still 95.6; we return to this below.

7.4 Results at $(3, 3)$

At $(3, 3)$ the problem is the same as at $(2, 3)$ apart from the value of n in its coefficients, and Table 5 and Figure 5 give the results. Where exact values exist, $k \leq 12$, the energy bounds again lie below them. The margin falls almost linearly in k , in steps that shrink from 0.98 to 0.85, and crosses zero between $k = 21$ and 24. The solves at $k \geq 24$, including the degrees next to half filling where BPS states may exist, are all feasible, as they must be if the method is sound. In the two decisive degrees, $k = 18$ and 21, the pure feasibility problem returns Farkas certificates whose dual semidefinite blocks are positive semidefinite, and the energy bounds there are positive as well.

k	$n(k)$	exact E_0	bound on E_0	t^*	$\ A^T y\ /\ y\ $	$b \cdot y/\ y\ $	verdict
3	27	6024	5457.3	-5.894			excluded
6	702	4634.9	4139.9	-4.912			excluded
9	18 981	3546.0	2824.4	-3.956			excluded
12	460 539	2472.0	1511.2	-3.012			excluded
15	8.7×10^6	—	598.2	-2.102			excluded
18	1.2×10^8	—	218.6	-1.219	2.3×10^{-11}	-4.0×10^{-3}	excluded
21	1.2×10^9	—	21.8	-0.374	6.1×10^{-12}	-1.2×10^{-3}	excluded
24	8.4×10^9	—	0	0	no certificate		not excluded
27–39	up to 1.1×10^{12}	—	0	0			not excluded

Table 5: The level-2 singlet bootstrap at (3, 3), seed-3 couplings, $E_{\text{vac}} = 6777$. Exact energies for $k \leq 12$ come from the Schur–Weyl basis. The two certificate columns characterise the Farkas vectors of the pure feasibility problem. The energy bound at $k = 21$ was obtained at reduced solver accuracy. By particle–hole symmetry the exclusions extend to $k \geq 60$.

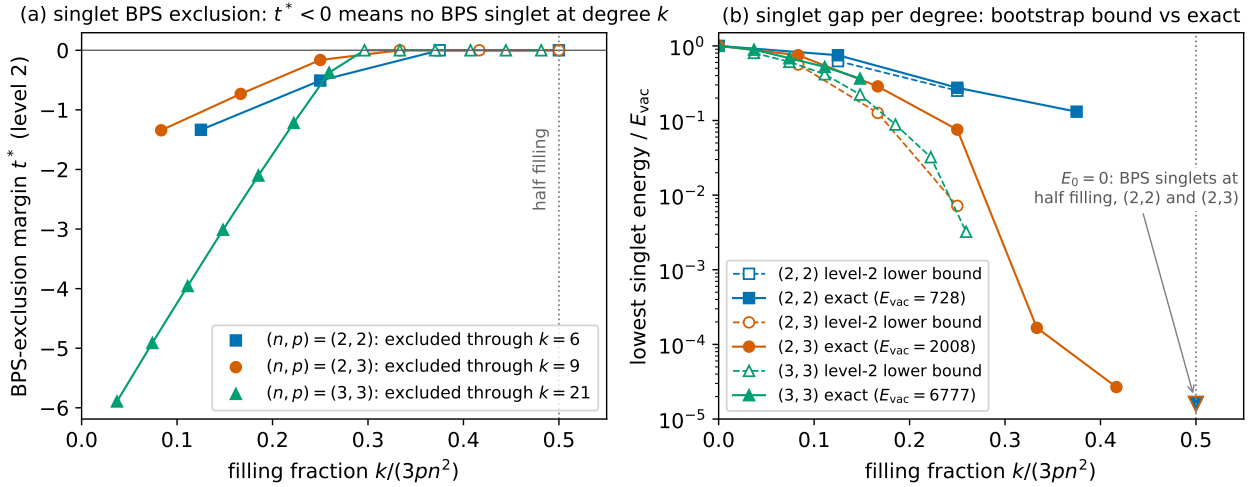


Figure 5: The singlet bootstrap at level 2. (a) The BPS-exclusion margin t^* against the filling fraction $k/3pn^2$ at (2, 2), (2, 3) and (3, 3); filled markers are excluded degrees ($t^* < 0$) and open markers are not. (b) The lowest singlet energy per degree in units of the vacuum energy $E_{\text{vac}} = n^3 \sum_{abc} C_{abc}^2$: bootstrap lower bounds (dashed, open markers, with vanishing bounds omitted) against the exact values (solid) where these are known; the triangles at half filling mark the BPS degrees at $n = 2$.

The certificates are robust in a precise sense. If a singlet BPS state existed, its functional x^* would satisfy $Ax^* + s = b$ with $s \in K$, so that $0 \leq \langle y, s \rangle = b \cdot y - (A^T y) \cdot x^*$. Every monomial of total length L obeys $|\phi(m)| \leq \|m\| \leq n^L$, with $L \leq 8$ here, so $\|x^*\| \leq 3^8 \sqrt{1.6 \times 10^5} \approx 2.6 \times 10^6$, and the right-hand side is at most $(-4.0 \times 10^{-3} + 6 \times 10^{-5})\|y\| < 0$ at $k = 18$. This is a contradiction with a factor of about 70 to spare, and the same holds at $k = 21$. The argument is computer assisted only through the floating-point evaluation of $A^T y$, $b \cdot y$ and the eigenvalues of the dual blocks, which we have not yet repeated in interval arithmetic.

Two conclusions follow. The first is fortuity at $p = 3$. The singlet cohomology of the (3, 3) model vanishes in degree 18, which is half filling at (2, 3), so by the argument of Section 4.3 none of the 1680 singlet BPS classes of the (2, 3) model has a rank-3 preimage. All of them are fortuitous in the ordinary sense, with the caveats of a numerical certificate and of the refined criterion of [14]. The second is partial concentration at rank three. There are no singlet BPS states at (3, 3) for $k \leq 21$ and, by particle–hole symmetry, none for $k \geq 60$. The degrees $24 \leq k \leq 36$ and their mirror images remain open, besides the middle pair $k = 39, 42$, which carries equal numbers of BPS states because $I_0(3, 3) = 0$, possibly none.

7.5 Where the bootstrap stops

Level 2 stops about half-way to half filling at all three points, much as it did for the adjoint models [7], and (2, 2) shows why. Diagonalising H on the full Fock sector $N_A = N_B = N_C = m$, with every gauge representation included, we find no BPS states at all for $m = 2$, whose lowest level is 70.9, but at least twelve for $m = 3$, all

property	$p = 2$	$p = 3$ (and $p \geq 3$)
BPS entropy	$\ln I_0 = 3n \ln 3 + O(\ln n)$, proved; not macroscopic	$\ln I_0 = F^*(p)n^2 + O(\ln n)$, $F^*(3) = 1.2373$, 41% of the singlet log-count (large-deviation level; sign proviso for odd p)
Concentration	(2, 2): all 90 singlet BPS states at $k = 12$, exact; (3, 2): all 1680 at $k = 27$, numerical	(2, 3): all 1680 at $k = 18$, numerical; (3, 3): none at $k \leq 21$ (bootstrap), open for $24 \leq k \leq 36$
Fortuity	(2, 2): all 90 classes, to all orders	(2, 3): all 1680 classes, via the (3, 3) certificate at $k = 18$
Non-BPS statistics	not random-matrix; hidden $\mathcal{N} = 4$ on singlets	Gaussian orthogonal, 10024 levels at (2, 3)
Near-BPS gap	none (≥ 95.6)	0.054 next to the BPS degree; Turiaci–Witten not yet testable
Supersymmetry	$\mathcal{N} = 4$ on singlets (numerical, $n = 2, 3$)	$\mathcal{N} = 2$
Large n	planar; exact index	planar; index saddle solved

Table 6: Desirable properties of the quiver for $p = 2$ and $p \geq 3$, with their status.

of them gauge non-singlets, since the singlet gap at $k = 9$ is 95.6. The frontier thus sits exactly at the onset of non-singlet BPS states in the sector. This is natural, because positivity alone cannot tell a singlet from the gauge average of a non-singlet: the matrix M in (12) is positive for any state, and gauge invariance enters only through the Casimir conditions. Accordingly those conditions barely affect the exclusions. Switching them off changes the margins at $k = 3$ and 6 from -1.334 and -0.508 to -1.327 and -0.505 , although it lowers the energy bounds from 455 and 183 to 293 and 63, so the exclusions obtained so far rule out BPS states in every gauge representation at once. None of the extensions we have tried moves the (2, 2) frontier: level 3, with Gram blocks of 26 words; Gauss-law conditions $\phi(Y \text{Tr}[WG_v]) = 0$, with G_v the $u(n)_v$ generator matrix and W an open word from v back to v ; the finite- n trace relations; and the statement that for a singlet ψ each vector $W_{ij}\psi$ lies in a definite gauge representation, so that $\sum_{ij} \langle W_a \psi | \hat{C}_x | W_b \psi \rangle$ is a fixed multiple of M_{ab} . The last of these strengthens the margin at $k = 6$ by half but leaves $k = 9$ feasible, and at (3, 3) it likewise leaves $k = 24$ feasible. Going further will need constraints that tie positivity to the gauge representation of the state much more tightly than Casimir conditions do, rather than more operators.

8 Prospects for the $p \geq 3$ quiver

Table 6 sets the two cases side by side. The contrast is sharp. The $p = 2$ quiver is an elegant and highly structured corner, with a hidden $\mathcal{N} = 4$ on singlets, an exact index and a spectrum that is not random-matrix like, but its BPS entropy is provably sub-macroscopic. The $p \geq 3$ quiver has every property we have been able to test at the level we would hope for: a singlet BPS entropy that is a finite fraction of the total at order n^2 , a concentrated BPS sector, chaotic non-BPS spectra, small gaps next to the BPS degree, and only $\mathcal{N} = 2$ supersymmetry, which is what the $\mathcal{N} = 2$ super-Schwarzian needs. The large- p limit adds a pleasing interpolation: the singlet BPS entropy per fermion rises from zero at $p = 2$ towards the SYK value at $p = \infty$, so the flavour number dials between a gauge-dominated matrix regime and an SYK-like one, with planar large- n control at every p .

The caveats are equally clear, and they define the programme. Much of this rests on $n = 2$, and concentration is the property most at risk as n grows: in the adjoint models it was the growth of the window with the rank that failed, and although its mechanism is absent here, nothing yet proves that another one does not take its place. At $p = 2$ concentration survives the step to $n = 3$. At $p = 3$ the bootstrap confines the rank-3 singlet BPS states to $24 \leq k \leq 57$, which is far from a proof, because the open degrees are precisely those nearest half filling, where a widening window would first appear. The near-BPS gaps at (2, 3) are suggestive, but at this size the neighbouring charge sectors differ too much in entropy for the Turiaci–Witten density to apply, so the super-Schwarzian question remains open. Fortuity at $p = 3$ is now established, at the level of a numerical certificate, through the (3, 3) bootstrap. And the model is planar rather than melonic, so at fixed p there is no analogue of the SYK Schwinger–Dyson solution; large- n dynamics has to come from the bootstrap, from other matrix-model methods, or from the large- p limit discussed below.

Several next steps follow, and the first observation is that concentration beyond $n = 2$ has to be settled without help from the index. Since Q raises the fermion number by three and every singlet has $k \equiv 0 \pmod 3$, no R-charge fugacity is compatible with the pairing of non-BPS states in the singlet sector: $\text{Tr}(-1)^k y^{k/3}$ is protected

only at $y = 1$. The singlet index is therefore the only protected quantity of the gauged model, and its sign, which always agrees with half filling, is all it can say about where the BPS states sit.

The exact methods have now reached their limit. The Schur–Weyl basis of Section 7 has delivered the $(3, 2)$ point, where the near-BPS gap and the multiplet statistics can also be followed from $n = 2$ to $n = 3$, albeit in the $p = 2$ corner. At $(2, 3)$, with 300 584 singlets at half filling, it can still supply the 1680 BPS vectors for a direct test of BPS chaos at $p = 3$. Beyond that the singlet sectors explode: at $(3, 3)$ they hold 1.2×10^8 states at $k = 18$ and 1.1×10^{12} at half filling, and 3×10^8 at half filling at $(2, 4)$.

At $p = 3$ and $n \geq 3$ the matrix bootstrap [15, 16, 17] has taken over, in the singlet form of Section 7, and it has delivered its first target, the certificate at $(3, 3)$ and $k = 18$ that makes the $(2, 3)$ classes fortuitous. The second target, certificates for $24 \leq k \leq 36$, which would confine the rank-3 BPS states to the two singlet degrees adjacent to half filling, is out of reach at level 2, and the diagnosis of Section 7 indicates that more operators alone will not reach it. What is needed are constraints that tie positivity to the gauge representation of the state more tightly than Casimir conditions do; whether Gram matrices of products of open words, resolved into gauge irreps, or a formulation that works inside the singlet sector from the start can achieve this is the next methodological question. The near-BPS gaps and their scaling with n , the actual super-Schwarzian test, remain the harder final target.

On the analytic side, the large- p limit is where the model should become SYK-like. The holonomy saddle realises precisely the $\mathbb{Z}_3$ twist that makes the index of $\mathcal{N} = 2$ SYK large, and at fixed n the $3pn^2$ fermions with random flavour couplings should be governed by the Schwinger–Dyson equations of [3], with a gauge correction of relative order $1/p$ since $\dim G / \dim V = 1/p$. Deriving the super-Schwarzian and the chaos exponent in this limit, and reproducing the exact large- p expansion of $F^*(p)$ in Table 2 as a check, would establish the desired dynamics in a controlled corner of the same model; that $F^*(p)$ is smooth for all $p > 2$ suggests that $p = 3$ lies in the same phase. Monte Carlo integration of (8), by thermodynamic integration in p from the exact point $p = 2$, would test the finite- n predictions at $n = 6$ –12, and the generalised monotones of [14] should be checked. Finally, the hidden $\mathcal{N} = 4$ of the $p = 2$ quiver deserves a derivation and a test at $n = 3$, both for its own sake and because a small $\mathcal{N} = 4$ super-Schwarzian would require an $SU(2)$ R-symmetry that we have not found.

In summary, the fermionic $U(n)^3$ quiver with $p \geq 3$ flavours is, to our knowledge, a new candidate for a matrix SYK model. It has macroscopic and, at the sizes we can reach, concentrated BPS entropy, chaotic multiplet spectra and near-BPS gaps; its rank-2 BPS classes are fortuitous at both $p = 2$ and $p = 3$; it reduces to an exactly solvable log gas at the level of the index; and its large- p limit makes contact with $\mathcal{N} = 2$ SYK. Whether concentration survives at large n is now the central question. It holds at $(3, 2)$, and at $(3, 3)$ the bootstrap excludes singlet BPS states from every degree except the twelve nearest half filling; sharper bootstrap constraints for those degrees and the large- p limit are the next steps.

A Methods and reproducibility

All computations are reproducible from the repository, with the parameters, seeds, outputs and memory use recorded in the result files. Cohomology at $(2, 2)$ uses exact ranks modulo primes. The $n = 2$ spectra use Weyl-reduced zero-weight sectors (`src/quiver_n2.py`) with the exact Casimir penalty, Lanczos (`scripts/quiver_singlet_zero_modes.py`), dense singlet bases (`scripts/quiver_singlet_spectrum.py`), the Casimir-polynomial projection for the 10 024-level pair (`scripts/quiver_singlet_pair_large.py`, float64, peak 8.6 GB) and the $\mathcal{N} = 4$ and symmetry analysis (`scripts/quiver_p2_n4.py`, `scripts/quiver_commutant.py`). The fortuity test is `scripts/run_quiver_fortuity.py`. The Schur–Weyl singlet basis is `src/quiver_singlet_basis.py` with `scripts/quiver_singlet_spectrum_sw.py`, tested by `tests/test_quiver_singlet_basis.py` (singlet dimensions, $Q^2 = 0$, and the stored $n = 2$ spectra). The bootstrap uses the quiver trace algebra `src/quiver_trace.py`, checked against explicit Fock-space operators by `tests/test_quiver_trace.py`, and the semidefinite program `src/quiver_bootstrap.py` with `scripts/quiver_bootstrap_run.py`; `tests/test_quiver_bootstrap.py` evaluates every constraint on exact functionals. The full-sector zero modes are `scripts/quiver_sector_zero_modes.py`, and the bootstrap figure is `scripts/plot_quiver_bootstrap.py`. Singlet indices use `src/quiver_index.py` (dual Cauchy and the Littlewood–Richardson transfer matrix, exact integers). The log-gas analysis is `src/quiver_saddle.py` and `scripts/quiver_index_saddle.py`, tested by `tests/test_quiver_saddle.py` (reduced against full energies, Hessians against finite differences to 10^{-10} , the $p = 2$ formula, the continuum values, and the eigenvalue representation against direct quadrature at $n = 1$). The derivations are written out in `docs/derivations.md` (D18, D20 and D21) and the research log in `research/notes/quiver_project.md` (Sections 11 and 12 for the material of Section 7). The main data files

are results/data/quiver_singlet_index_dixon.json, quiver_singlet_zero_modes_n2_p3_k*.json, quiver_singlet_pair_n2_p3_k9_f64.json, quiver_p2_n4*.json, quiver_fortuity_n2_p2_k12.json, quiver_index_saddle.json, quiver_singlet_sw_n3_p2_seed3.json, quiver_singlet_sw_n3_p3_seed3.json, quiver_bootstrap_n*_p*_L*.jsonl and quiver_bootstrap_frontier_diagnostics.txt. The bootstrap runs used the seed-3 integer couplings and the interior-point solver Clarabel 0.11.1, with peak memory 3–4.5 GB.

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